



**University of Colombo, Sri Lanka**

*University of Colombo School of Computing*

**BACHELOR OF SCIENCE IN COMPUTER SCIENCE**

First Year Examination — Semester I— UCSC AY20 [held in October 2023]

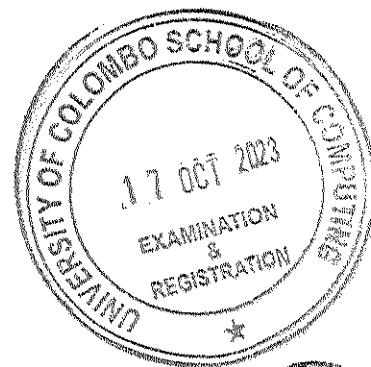
**SCS1207 — Software Engineering I**

(Two (2) Hours)

Answer all 4 questions

Number of Pages = 12

Number of Questions = 4



To be completed by the candidate

Index Number

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**Important Instructions to candidates:**

- Students should answer in the medium of English language only using the space provided in this question paper.
- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Write your index number **CLEARLY** on each and every page of the question paper.
- This paper consists of 4 questions in 12 pages (including the Cover Page).
- Answer **ALL** questions.
- Questions 1, 2, 3, 4 contains 20, 25, 25, 30 marks respectively.
- Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.
- Do not tear off any part of this Question Paper. Under no circumstances may this paper, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

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1. This question consists of 10 Multiple Choice Questions. **Underline the correct answer.** Each question contains **only one correct answer**.

[20 marks]

- (a). What is the main objective of requirements elicitation in software engineering?
- 1) To gather and understand developer needs and constraints.
  - 2) To prepare the system requirements document.
  - 3) To design system models.
  - 4) To gather and understand user needs and constraints.
  - 5) To define the project budget.
- (b). Which type of requirement describes what the system should do in response to specific inputs or stimuli?
- 1) Non-functional requirement
  - 2) Functional requirement
  - 3) High-level requirement
  - 4) Design requirement
  - 5) System requirement
- (c). What is the primary goal of requirements prioritization in software development?
- 1) To rank requirements based on their technical complexity.
  - 2) To ensure that all requirements are implemented in the first release.
  - 3) To allocate resources effectively and deliver the most valuable features early.
  - 4) To determine the project budget.
  - 5) To create a detailed system architecture.
- (d). What is the purpose of requirements validation in the software development process?
- 1) To ensure that requirements are complete, consistent, and correct.
  - 2) To ensure that user feedbacks are documented properly.
  - 3) To prioritize and design major functional requirements.
  - 4) To ensure that requirements are designed properly.
  - 5) To ensure that requirements are aligned with the test cases.
- (e). Which UML diagram is used to describe the interactions between different objects in a system over time, typically in response to events or messages?
- 1) State chart diagram
  - 2) Activity diagram
  - 3) Sequence diagram
  - 4) Component diagram
  - 5) Deployment diagram

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- (f). In a Class diagram, what does an association line between two classes with a "1" and a "0..1" mean?
- 1) One class inherits zero or one instances of the other.
  - 2) One class inherits zero or one subclasses of the other.
  - 3) One class has a one-to-many relationship with the other.
  - 4) One class has a one-to-one relationship with the other.
  - 5) There is no relationship between the classes.
- (g). In UML, what does a black diamond shape with a solid line (  ) connecting two classes in a Class diagram represent?
- 1) Inheritance
  - 2) Aggregation
  - 3) Composition
  - 4) Association
  - 5) Dependency
- (h). In the 4+1 View Model architecture, which view focuses on capturing the system's dynamic behavior, showing how various components interact over time?
- 1) Logical View
  - 2) Process View
  - 3) Development View
  - 4) Deployment View
  - 5) Use Case View
- (i). In a client-server architectural pattern, what role does the 'server' typically play?
- 1) Handling user interactions and presenting the user interface.
  - 2) Acting as an intermediary between clients.
  - 3) Initiating communication with clients.
  - 4) Managing data access and storage.
  - 5) Managing client devices and hardware.
- (j). Which architectural pattern is known for facilitating the separation of an application's presentation logic, business logic, and data access logic?
- 1) Repository pattern
  - 2) Model-View-Controller (MVC)
  - 3) Observer pattern
  - 4) Client-server pattern
  - 5) Layered architecture

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2. (a). Briefly justify the distinction between Software Development Projects and other types of projects by providing **at least five (05) factors**.

[5 marks]

|    |
|----|
| 1. |
| 2. |
| 3. |
| 4. |
| 5. |

- (b). Underline whether the following statements are TRUE or FALSE. Justify your answer in the space provided.

[10 marks]

**Note: No marks will be provided, only for mentioning the statement is True/False. In either case, a proper justification is required.**

- i. Software is simply another term used to refer to Computer Programs. [TRUE/FALSE]

***Justification HERE:***

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- ii. Software Specification, Software Design, and Software Development are the ONLY fundamental activities relevant to the Software Engineering Life Cycle. [TRUE/FALSE]

*Justification HERE:*

- iii. The total cost of a software development project is the sum of the effort cost of paying software developers. [TRUE/FALSE]

*Justification HERE:*

- iv. Software Engineering is a profession that ONLY applies theories of Computer Science to develop Customized Software. [TRUE/FALSE]

*Justification HERE:*

- v. LEAN and Kanban are two examples of a Plan-driven Software Development Approach. [TRUE/FALSE]

*Justification HERE:*

- (c). Consider the following definitions related to the SCRUM terminology.
- (A) The sole person responsible for managing the Product Backlog.
  - (B) A dynamic list of items that may be added, removed, or reprioritized based on the progress.
  - (C) A development iteration that is usually 2 to 4 weeks long.
  - (D) A daily meeting of the Scrum team that reviews progress and prioritizes work to be done on that day.

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- (E) Responsible for delivering the work through the sprint.
- (F) A piece of work that is potentially shippable.
- (G) A collaborative meeting takes place at the end of each iteration to focus on the product and its functionality.
- (H) Responsible for interfacing with the rest of the company and ensuring the Scrum team is not diverted by outside interference.
- (I) A collaborative meeting held at the beginning of each iteration to define the work that will be undertaken during the iteration and to set clear goals and expectations.
- (J) An inspect-and-adapt cycle to enable the team to continuously improve its processes and performance incrementally over time.
- (K) Ordered list of all features, functions, requirements, enhancements, and fixes that constitute the changes to be made to the product in future releases.

Identify the appropriate SCRUM terminology for the above definitions and categorize them into Roles, Artifacts, and Ceremonies within the SCRUM Framework. Specify the Letter of each statement and the relevant SCRUM terminology in the appropriate space.

For example: [A] - Product Owner

[10 marks]

| SCRUM ROLES                | SCRUM ARTIFACTS | SCRUM CEREMONIES |
|----------------------------|-----------------|------------------|
| <i>[A] - Product Owner</i> |                 |                  |
|                            |                 |                  |
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3. (a). Briefly explain the **Test-First Development (TFD)** concept and justify its importance in modern software development practices by at least providing **two (02)** factors.

[3 marks]

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- (b). Compare and contrast traditional Waterfall Testing over Agile Testing.

[10 marks]

|              | Waterfall Testing | Agile Testing |
|--------------|-------------------|---------------|
| Similarities |                   |               |
|              |                   |               |
| Differences  |                   |               |
|              |                   |               |
|              |                   |               |

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- (c). Match the statements in Column I with the most suitable testing terminology from the list of words provided and write the answer in Column II.

[6 marks]

[List of Words: Static Testing, Structural Testing, Regression Testing, Development Testing, Release testing, Stress Testing, Beta Testing, Dynamic Testing, Jera Testing, Test First Development, Test Box Testing]

| COLUMN I |                                                                                                                                                                                      | COLUMN II - ANSWER |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| (i)      | Gather real-world user feedback to assess the software's performance and identify any remaining issues or defects.                                                                   |                    |
| (ii)     | A type of testing that involves executing the software and observing its behavior, performance, and output in various scenarios to identify defects, errors, or unexpected behavior. |                    |
| (iii)    | The primary purpose is to ensure that the recent modifications have not adversely affected the software's existing functionality and that it continues to operate correctly.         |                    |
| (iv)     | The main goal is to identify the system's breaking points, weaknesses, and vulnerabilities when subjected to a heavy load, high traffic, or resource-intensive scenarios.            |                    |
| (v)      | A testing stage where all testing activities are carried out by the team developing the system.                                                                                      |                    |
| (vi)     | A method of software testing that examines the functionality of an application by peering into its internal structures or workings of an application.                                |                    |



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(d). There will be negative marks for incorrect responses to avoid guessing for (d)-i to (d)-iii.

[6 marks]

- i. Which of the following activities is/are associated with the software re-engineering processes?

| Activity                      | Tick (✓) the relevant activity |
|-------------------------------|--------------------------------|
| Impact Analysis               |                                |
| Reverse Engineering           |                                |
| Data Clumping                 |                                |
| Program Structure improvement |                                |
| Architecture Reengineering    |                                |

- ii. Which of the following statements is/are correct regarding Legacy Systems?

| Statement                                                                                                     | Tick ( ✓ ) the correct statement |
|---------------------------------------------------------------------------------------------------------------|----------------------------------|
| Legacy system replacement is risky due to the tight integration of business processes and policies.           |                                  |
| Organizations often hesitate to replace legacy systems as they are typically more secure than modern systems. |                                  |
| Replacing legacy systems with an entirely new one is always recommended as they use old technologies.         |                                  |
| Legacy Systems cannot be transformed to improve their maintainability by applying re-engineering techniques.  |                                  |

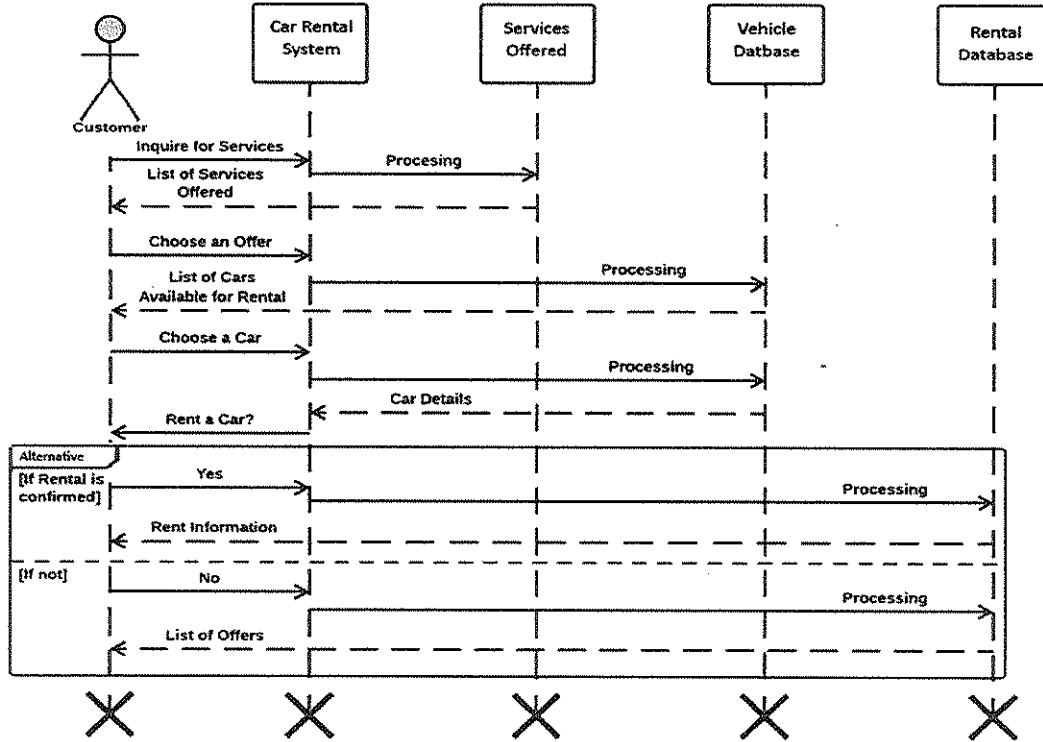
- iii. Which of the following testing techniques is/are suitable for the Acceptance testing phase?

| Testing Technique       | Tick ( ✓ ) the relevant technique |
|-------------------------|-----------------------------------|
| Black Box               |                                   |
| White Box               |                                   |
| Stress Testing          |                                   |
| Inspections and Reviews |                                   |
| Prototyping             |                                   |

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4. (a). Consider the following sequence diagram.



Explain the above sequence diagram by clearly describing the involving actors and objects, all messages and message types. Indicate any assumptions if any.

[10 marks]

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(b). AeroTravel a leading airline ticket booking company who plans to provide online services to their customers. Using this application, a passenger should be able to search for airlines, check for availability of flights, book tickets and make payments. When booking a ticket, they should have the option to select preferred seats. When making a payment a card validation should happen. System admins would be able to manage flight schedules and cancel booked tickets if required. Passengers should also be able to check the status of the flights through the system.

- i. Draw a use case diagram for the above case study clearly illustrating the associations between the actors and use cases. Clearly show the extend and include relationships between the use cases if there are any.

[10 marks]

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- ii. Suppose that 'Booking' is a class of the above system. Identify **three (3)** other possible classes.

[3 marks]

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- iii. Suppose the 'Booking' class represents a booked flight reservation made by a passenger. Identify possible attributes and operations relevant to the Booking class providing a description of each.

For example: bookingID: Unique identifier for each booking

[7 marks]

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